Disentangling Evaluation Distortions in 2D-to-3D Brain Tumor MRI Segmentation: A Patient-Grouped Leakage, Calibration, and Volumetric Post-Processing Audit on TCGA-LGG

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- **Author Contributions:** Shayan Ganji Nataj conceptualized the study, developed the methodology and the CAL-3DH algorithm, conducted formal analysis and software implementation, executed the experiments, curated and validated data, and drafted the original manuscript. Keivan Borna supervised the research project, contributed to methodological formulation, provided critical intellectual input and resources, and reviewed and edited the manuscript. All authors read and approved the final manuscript.
- **Ethics Approval:** This retrospective study was conducted using publicly available, fully de-identified human neuroimaging and genomic data from The Cancer Genome Atlas (TCGA) and The Cancer Imaging Archive (TCIA). In accordance with institutional policy and the 1964 Declaration of Helsinki, this secondary computational audit of publicly available, fully de-identified TCGA/TCIA datasets did not involve new human-subject data collection or clinical intervention.
- **Consent to Participate and Consent to Publish:** As documented by the TCGA and TCIA consortia [18, 19], all human subjects data were collected by participating tissue source sites under institutional review board approvals and patient informed consent policies per TCGA/TCIA governance. All MRI scans and clinical records are fully anonymized and HIPAA-compliant; no identifiable patient information is included in this article.
- **Data and Code Availability:** The imaging data utilized in this study are publicly accessible through The Cancer Imaging Archive (TCIA; Clark et al., 2013; Pedano et al., 2016) via the persistent DOI: `https://doi.org/10.7937/K9/TCIA.2016.L4LTD3TK` and mirrored on Kaggle at `https://www.kaggle.com/datasets/mateuszbuda/lgg-mri-segmentation`. Matched clinical and genomic records were extracted from the companion table `data.csv` distributed with the benchmark dataset (Buda et al., 2019; Pedano et al., 2016), originally curated from the TCGA pan-glioma genomic study (Ceccarelli et al., 2016). All source code, reproduction scripts, and 5-fold cross-validation pipelines are openly maintained at GitHub: `https://github.com/Shayan-Ganji/lgg_segmentation`.